

YOSHEE JAIN

Undergraduate Research Assistant
 Siebel School of Computing and Data Science
yosheej2@illinois.edu — yosheejain.com

Education

University of Illinois Urbana-Champaign

Champaign-Urbana, IL

- Bachelor of Science in Computer Science
- Minor in Statistics
- Completed Data Science Certification
- **Thesis:** Towards Identifying Domain-Specific Programming Plans at Scale: Needs, Challenges, and Solutions

May 2026

GPA: 4.0/4.0

Research Objective

My research interests lie in designing interactive learning environments that help people from diverse backgrounds learn and use computing. I understand how students anticipate using computational knowledge in their future careers using HCI methods. I then leverage learning science research to theorize and prototype pedagogical scaffolds that meaningfully support learners' goals. Finally, using AI/ML techniques, I develop and evaluate human-centered systems that implement these instructional interventions at scale.

Topics: Human-Centered AI, Human-Computer Interaction, CS Education, Human-AI Collaboration, Cognitive Science

Publications

Preprints

* Indicates Equal Contribution.

- C₄ 2026
arXiv **Y. Jain**, H. Do, Z. Wu, A.Y. Wang. Exploring the Role of Tracing in AI-Supported Planning for Algorithmic Reasoning. *Submitted Feb 2026.*

Heavily Peer-Reviewed Conference Proceedings

- C₃ 2026
ICWSM A. Goyal, C. Lambert, **Y. Jain**, E. Chandrasekharan. Uncovering the Internet's Hidden Values: An Empirical Study of Desirable Behavior Using Highly-Upvoted Content on Reddit. *2026 International AAAI Conference on Web and Social Media.*
- C₂ 2025
CHI **Y. Jain***, M. Demirtas*, K. Cunningham. PLAID: Supporting Computing Instructors to Identify Domain-Specific Programming Plans at Scale. *2025 ACM CHI Conference on Human Factors in Computing Systems.*
- C₁ 2025
HCII **Y. Jain**, J. Hollander, A. He, S. Tang, L. Zhang, J. Sabatini. Exploring the Potential of Large Language Models for Estimating the Reading Comprehension Question Difficulty. *2025 International Conference on Human-Computer Interaction.*

Journal Articles

- J₃ 2025
NPJ AI K. Saha, **Y. Jain**, M. De Choudhury. Linguistic Comparison of AI- and Human-Written Responses to Online Mental Health Queries. *NPJ Artificial Intelligence* 2026.
- J₂ 2025
JMIR Aging S. Kaliappan, C. Liu, **Y. Jain**, R. Karkar, K. Saha. Online Communities as a Support System for Alzheimer's and Dementia Care: Large-scale Exploratory Study. *JMIR Aging* 2025.
- J₁ 2025
ACM Health K. Saha, **Y. Jain**, C. Liu, S. Kaliappan, R. Karkar. AI vs Humans for Online Support: Comparing the Language of Responses from LLMs and Online Communities of Alzheimer's Disease. *ACM Transactions on Computing for Healthcare* 2025.

Posters

- P₃ 2025
CSCW E. Mok*, **Y. Jain***, S. Gottiparthi*, E. Chandrasekharan. Voices Lost in the Algorithm: Improving Instagram's Moderation of Mental Health Communities. *2025 ACM Conference on Computer-Supported Cooperative Work and Social Computing.*
- P₂ 2024
CSCW C. Lambert, **Y. Jain**, K. Saha, E. Chandrasekharan. Investigating How Gilds Were Employed On Reddit. *2024 ACM Conference on Computer-Supported Cooperative Work and Social Computing.*
- P₁ 2023
ICER **Y. Jain**, K. Cunningham. Towards Methods for Identifying High-Quality Domain-Specific Programming Patterns. *2023 ACM Conference on International Computing Education Research V.2.*

Symposiums & Lightly-Peer Reviewed Conferences

S ₂ 2025 TAPIA	Y. Jain, K. Cunningham. Exploring the Use of LLMs to Generate Contextualized Worked Examples From Programming Patterns. <i>TAPIA Conference 2025</i>
S ₁ 2025 NW	Y. Jain, M. Demirtas, K. Cunningham. PLAID: Supporting Computing Instructors to Identify Domain-Specific Programming Plans at Scale. <i>AI, Education, and the Learning Sciences, Northwestern University.</i>

Thesis

T ₁ 2024 BS	Y. Jain. Towards Identifying Domain-Specific Programming Plans at Scale: Needs, Challenges, and Solutions. <i>Bachelor's Thesis.</i>
---------------------------	---

Research Experience

- **Undergraduate Research Assistant** **Jan 2023 – Present**
Dr. Katie Cunningham *University of Illinois Urbana-Champaign*
CS Education, Human-Computer Interaction, AI, Experimental Research, Mixed-Methods Research
- **Undergraduate Research Assistant** **Jan 2026 – Present**
Dr. Max Fowler *University of Illinois Urbana-Champaign*
CS Education, AI, Quantitative Analysis, Natural-Language Processing, Statistical Modeling
- **Undergraduate Summer Research Assistant** **May 2025 – Feb 2026**
Dr. April Yi Wang, Dr. Zihan Wu, Dr. Heejin Do *Federal Institute of Technology Zurich (ETH Zürich)*
Human-Computer Interaction, AI, CS Education, Experimental Research, Mixed-Methods Research
- **Undergraduate Summer Research Intern** **Jun 2025 – Dec 2025**
Dr. Juho Kim, Dr. Seungju Kim, Dr. Saelyne Yang, Chen Zhou *Korea Advanced Institute of Science and Technology*
Human-Computer Interaction, Computer Vision, Multimodal LLMs, Education
- **Undergraduate Research Assistant** **Aug 2023 – Aug 2025**
Dr. Eshwar Chandrasekharan, Charlotte Lambert *University of Illinois Urbana-Champaign*
Human-Computer Interaction, Social Computing, Data Science, Causal Inference
- **Undergraduate Research Assistant** **Aug 2023 – May 2025**
Dr. Koustuv Saha, Dr. Ravi Karkar *University of Illinois Urbana-Champaign*
Human-Computer Interaction, Social Computing, Data Science, AI, Mixed-Methods Research
- **Undergraduate Research Intern** **May 2024 – Aug 2024**
Dr. Jionghao Lin, Dr. John Sabatini, Dr. John Hollander, Dr. Liang Zhang *Carnegie Mellon University*
Human-Computer Interaction, AI, Education

Honors, Awards, Fellowships, and Scholarships

• Bronze Tablet	2026
• Distinguished Chancellor's Scholar	2026
• Finalist in the ACM Undergraduate Student Research Competition	2025
• Outstanding Course Assistant Award	2025
• Jeffrey P. Blahut Memorial Scholarship	2025
• CHP Outstanding Senior Award	2025
• John and Dorothy Durkin Women in Engineering Scholarship	2025
• Harold and Ruth Hayward/Tau Beta Pi Award Fund	2025
• Illinois Engineering Achievement Scholarship	2024-2025
• Illinois Engineering Outstanding Scholarship	2024-2025
• Conference Travel Grants	2025
• Dean's List	2022–Present
• Summer Research Award	2024
• Distinguished Undergraduate Researcher Certificate	2024
• Distinguished Active Member, Tau Beta Pi (The Engineering Honor Society)	2024
• Chancellor's Scholar	2022-Present
• Edmund J. James Scholar	2022-Present

Teaching Experience

- **Course Assistant**, CS 101: Intro Computing: Engrg & Sci. Fall 2025, Spring 2026
- **Course Assistant**, CS 102: Little Bits to Big Ideas Spring 2025
- **Project Manager**, CS 124 Honors: Introduction to Computer Science I Spring 2023

Service

Conference Student Volunteering

- ACM SIGCSE (2026)

University of Illinois, Urbana-Champaign

- CS Peer Mentor Jan 2026 - Present
- CS Student Ambassadors/Research Scholar Jan 2023 - Present

Skills

Languages: Python, C/C++, Java, $\LaTeX$, R, HTML/CSS, JavaScript, Flask, Django, TypeScript, React * Taken at UIUC.

Frameworks: Cursor, Visual Studio Code, Git/GitHub, Android Studio, PyCharm, REST APIs

Research Methods: System Design, Qualitative Analysis, Quantitative Research, Mixed Methods Analysis, User Interviews, Quasi-Experimental Studies, and Causal Inference Analysis

Relevant Courses*: Methodological Pluralism (Insight into HCI Methodologies), Computer Science Education Research, Computational Social Science, Statistics and Probability, Statistical Programming (R), Database Systems, Text Mining, User Interface Design, Advanced Data Analysis, Algorithms and Models of Computation, Applied Machine Learning